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RESEARCH ARTICLE



A movement-aware measure for trajectory similarity and its application for ride-sharing path extraction in a road network

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ABSTRACT

Recognizing common travel paths of crowds in a road network is valuable for understanding human mobility patterns and developing intelligent ride-sharing services. To achieve this, it is critical to measure the similarity of their trajectories. Although many measures have been proposed in the past decades, they often ignore movement consistency, exhibit one or more deficiencies in the face of noise and misaligned trajectories, or require extra parameters to tune predictions. In this paper, we propose an improved similarity measure called the directed segment path distance (DSPD), which considers the spatial proximity and movement consistency of trajectories. By integrating the spatial proximity distance and moving direction similarity between trajectories, the DSPD is a competitive parameter-free similarity measure that can effectively distinguish trajectories with different movement characteristics. To verify the effectiveness of the DSPD, we conducted a quantitative comparative study between the DSPD measure and 11 state-of-the-art trajectory similarity measures on six simulated trajectory datasets and applied the DSPD to two typical application scenarios: trajectory clustering for road network generation and retrieving common trajectories for ride-sharing path planning. The results demonstrate the effectiveness, robustness, and superiority of the DSPD and its great potential in trajectory search, clustering, and classification.

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Trajectory similarity measure; trajectory clustering; directional consistency; ride-sharing path extraction; GPS trajectory

1. Introduction

Measuring the similarity of trajectories is fundamental for mobility analysis and applications such as trajectory clustering (Chen *et al.* 2023, Zhang *et al.* 2006), movement pattern mining (Jin *et al.* 2020, Park *et al.* 2021), anomaly detection (Wang *et al.* 2021), road network generation (Deng *et al.* 2018, Sheng and Yin 2018) and user profile recognition (González *et al.* 2008). Measuring trajectory similarity is the first step in

searching for and grouping similar trajectories, and the application of different similarity measures may significantly influence the results (Besse *et al.* 2016, Su *et al.* 2020, Yuan *et al.* 2017). Therefore, it is crucial to select an appropriate trajectory similarity measure based on specific tasks. For example, when applying trajectory clustering methods to extract urban roads, the trajectory direction must be incorporated into the calculation of trajectory similarity (Deng *et al.* 2018). This consideration allows for the differentiation of roads in distinct directions even when they are in close spatial proximity. In addition, with the development of advanced techniques such as the Internet of Things and Global Navigation Satellite Systems, massive individual and vehicle trajectories have been collected and accumulated (Kong *et al.* 2018). Crowdsourced trajectory datasets provide significant potential opportunities for urban research (Al-Dohuki *et al.* 2019, Liu *et al.* 2022) and human mobility analysis (Kan *et al.* 2019, Peng *et al.* 2023). In these analyses, trajectory similarity measurement is a key step and a fundamental operation. However, measuring trajectory similarity is difficult and has several challenges.

1. Crowdsourced trajectories often vary in length and sampling rate. An effective trajectory similarity measure should be able to handle trajectories of different lengths or sampling rates without requiring additional trajectory alignment operations.
2. A reliable trajectory similarity measure should be insensitive to noise points in trajectories. This is the basis for ensuring the reliability of the trajectory similarity calculation results.
3. To enhance the usability and reduce the influence of user subjectivity on the similarity evaluation results, the trajectory similarity measure should not require additional parameters. In particular, for trajectory data with uneven density distribution and without prior user-knowledge, determining an appropriate parameter can be challenging for users and it may generate different results using different parameter values.

Although numerous trajectory similarity measures have been proposed in recent decades, such as the Hausdorff and Fréchet distances, dynamic time warping (DTW), and longest common sequence (LCSS), these measures often exhibit one or more deficiencies in solving the above challenges (Hu *et al.* 2023, Su *et al.* 2020, Tao *et al.* 2021). Moreover, most existing measures have been developed based on proximity or shape similarity, but they do not consider the consistency of the target movement. Therefore, this paper proposes a movement-aware measure for trajectory similarity called the directed segment path distance (DSPD) to address these challenges. The DSPD is a powerful parameter-free similarity measure that can effectively distinguish trajectories with different movement characteristics. The proposed measure is an extension and supplement to existing trajectory similarity measures and has important potential in trajectory data clustering and GIS applications, such as road map generation and human mobility pattern detection.

The remainder of this paper is organized as follows: [Section 2](#) provides a review of existing studies. [Section 3](#) introduces the details of the proposed trajectory similarity

measure. The experimental results for the simulated datasets and real-world trajectories are presented in [Sections 4](#) and [5](#), respectively. In [Section 6](#), the robustness of the proposed method is discussed. [Section 7](#) summarizes the study and presents its limitations and possible directions for future research.

2. Related work

In recent years, many measures have been proposed to evaluate trajectory similarity in the literature. For example, Magdy *et al.* (2015) compared trajectory similarity measures based on geometric shapes and spatiotemporal properties. Hu *et al.* (2023) categorized trajectory similarity measures into free-space- and network-constraint-based measures. Su *et al.* (2020) distinguished between discrete and continuous measures and Besse *et al.* (2016) compared shape- and warping-based trajectory similarity measures. Additional contributions include the studies by Wang *et al.* (2013) and Sousa *et al.* (2020). Based on existing studies, this study reviews trajectory similarity measures for trajectory clustering and analyzes the advantages and disadvantages of the current representative trajectory similarity measures. Trajectories are typically modeled as either time sequences or spatial lines. Given this, the existing trajectory similarity measures were roughly divided into two main categories: sequence-based and shape-based measures.

Sequence-based measures calculate the trajectory similarity in the order of the trajectory points. Euclidean, Manhattan, and Mahalanobis distances, as well as the vector similarity evaluation model (VSEM), are well-known sequence-based measures for their computational simplicity and efficiency (Furtado *et al.* 2018, Pathak *et al.* 2006, Wang *et al.* 2013, Yang *et al.* 2018). However, because these measures directly calculate point-to-point distances between trajectories, they are sensitive to noise points and require trajectories of the same length. In contrast, DTW (Berndt and Clifford 1994), LCSS (Vlachos *et al.* 2002), edit distance on real sequence (EDR) (Chen *et al.* 2005), and their variants (Deng *et al.* 2018, Sousa *et al.* 2020, Tao *et al.* 2021), can handle unequal-length trajectories by allowing nonlinear sequence alignment. These measures are widely used in speech recognition and time series clustering (Dilmi *et al.* 2020, Yadav and Alam 2018). Among these measures, the LCSS is less sensitive to noise and uneven sampling rates but requires an extra distance parameter (Vlachos *et al.* 2002). The LCSS quantifies the trajectory similarity by identifying the longest common subsequence. Deng *et al.* (2018) proposed an improved LCSS distance, called the longest common sub-track (LCST), by explicitly considering direction consistency. The LCSS and LCST have been used in trajectory clustering for movement pattern detection (Choong *et al.* 2017, Widyantara *et al.* 2023). However, both measures require manually setting a matching distance threshold, and a small change in the distance threshold may lead to completely different results. For trajectory data with an uneven density distribution, it is usually difficult to determine an appropriate distance threshold without prior knowledge. The EDR and related methods, such as Edit Distance with Projections (EDwP), are commonly used in graph matching (Fischer *et al.* 2015), trajectory compression (Makris *et al.* 2021), and anomaly identification (Lan and Yoon 2022).

Similar to the LCSS, the EDR, and EDwP are also limited by distance threshold selection (Ranu *et al.* 2015).

Shape-based measures focus on the geometric features of trajectories, emphasizing the overall consistency of the shape and structure rather than the point sequence (Besse *et al.* 2016, Lin and Su 2008, Zhang and Shi 2021). The Fréchet distance, proposed by Alt and Godau (1995), takes the infimum of maximum distances between two curves to evaluate curve similarity. Its variants include discrete Fréchet (DF) (Eiter and Mannila 1994) and weak Fréchet (Bogacki *et al.* 1997). They are useful for analyzing co-moving objects, aiding in trajectory retrieval (Gudmundsson *et al.* 2023), and sequence similarity comparison (VanderKraats *et al.* 2013). Conversely, the Hausdorff distance (HD) disregards the point order and directly measures the geometric variations between point sets by determining the maximum of the minimum distances between points, which is commonly used in matching and clustering polygons or polyline objects (Chen *et al.* 2011, Kim *et al.* 2023). Despite their popularity, Fréchet and Hausdorff distances are sensitive to noise. To reduce the impact of noise, Lin and Su (2008) proposed the one-way distance (OWD). Its symmetric version (SOWD) averages two OWDs between two trajectories. Building on the OWD, the symmetrized segment-path distance (SSPD) proposed by Besse *et al.* (2016) calculates the projected distances from points to the line segments of another trajectory. The SSPD enhances noise robustness without parameter dependence but ignores the directional difference of the trajectories. Zhang and Shi (2021) proposed the SSPD* to identify trajectory groups with similar directions. However, OWD, SSPD, and SSPD* may not accurately estimate the similarity between a trajectory and its parts (i.e. sub-trajectories) like LCSS, to identify groups of objects with similar movement patterns. Based on the above review and analysis, a summary and comparison of existing representative trajectory similarity measures are presented in Table 1.

To solve the limitations of existing measures, this study defines an improved parameter-free similarity measure that considers the movement consistency of trajectories and demonstrates its applications in road network generation and common trajectory retrieval for ride-sharing path planning.

3. Movement-aware trajectory similarity measure (DSPD)

3.1. Preliminaries

Inspired by the SSPD measure (Besse *et al.* 2016), this study develops an improved shape-based trajectory similarity measure called the DSPD. Before introducing the DSPD measure, we formally define the basic definitions related to the trajectory similarity measure.

Definition 1 (Trajectory): A Trajectory $T = \{p_1, p_2, \dots, p_n\}$ is defined as a time-ordered sequence of spatial points that records the historical locations of a moving object, where n is the number of points in the trajectory.

Definition 2 (Trajectory segment): The trajectory segment s_i is defined as a segment line that connects two adjacent points p_i and p_{i+1} in T . Let T_s be the set of trajectory

Table 1. Comparison of different trajectory similarity measures.

Category	Measure	Parameter free	Noise insensitivity	Unequal length trajectories	Compatible with sub-trajectories	Trajectory direction	Symmetric	Time complexity
Sequence-based measures	ED	✓					✓	$O(n + m)$
	DTW	✓		✓			✓	$O(nm)$
	LCSS		✓	✓	✓		✓	$O(nm)$
	LCST		✓	✓	✓	✓	✓	$O(nm)$
	EDR			✓			✓	$O(nm)$
	EDwP	✓		✓			✓	$O(nm)$
	VSEM					✓	✓	$O(n + m)$
Shape-based measures	Fréchet	✓		✓			✓	$O(nm \log(nm))$
	DF	✓		✓			✓	$O(nm)$
	Hausdorff	✓		✓			✓	$O(nm)$
	OWD	✓	✓	✓			✓	$O(nm)$
	SOWD	✓	✓	✓			✓	$O(nm)$
	SSPD	✓	✓	✓			✓	$O(nm)$
	SSPD*	✓	✓	✓		✓	✓	$O(nm)$
	DSPD	✓	✓	✓	✓	✓	✓	$O(nm)$

* n and m are the numbers of sampling points of the two trajectories and ✓ is satisfied.

DF: discrete Fréchet; DSPD: Directed SegmentPath Distance; DTW: Dynamic Time Warping; ED: Euclidean Distance; EDR: Edit Distance on Real Sequences; EDwP: Edit Distance with Projections; LCSS: Longest Common Sequence; LCST: Longest Common Sub-Track; OWD: One Way Distance; SOWD: symmetric OWD; SSPD: Symmetrized Segment-Path Distance; VSEM: vector similarity evaluation model.

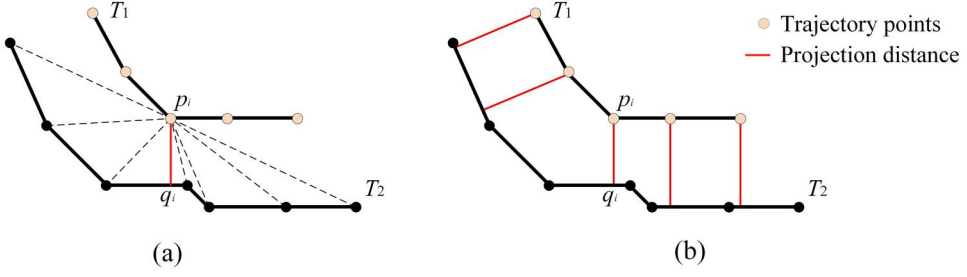


Figure 1. Illustrations of the projected distance and segment-path distance of (a) projection distance from p_i to T_2 and (b) segment-path distance of T_1 to T_2 .

segments of T , denoted by $T_s = \{s_1, s_2, \dots, s_{n-1}\}$. The length of T_s equals the sum of the lengths of all segments, i.e. $|T_s| = \sum_{i=1}^{n-1} |s_i|$.

Definition 3 (*Point to segment distance*): Point to segment distance (PSD) is the shortest distance from a point to a trajectory segment. As shown in Figure 1a, given two trajectories T_1 and T_2 , let p_{i1} be a trajectory point in T_1 , T_{s2} be the trajectory segment set of T_2 , s_{i2} be a trajectory segment in T_{s2} , and p_{j2} and $p_{(j+1)2}$ be the start and end points of s_{i2} . Subsequently, $d_{psd}(p_{i1}, s_{i2})$ is calculated as

$$d_{psd}(p_{i1}, s_{i2}) = \begin{cases} \|p_{i1}, p_{i1}^{\text{proj}}\| & \text{if } p_{i1}^{\text{proj}} \in s_{i2} \\ \min(\|p_{i1}, p_{j2}\|, \|p_{i1}, p_{(j+1)2}\|) & \text{otherwise} \end{cases} \quad (1)$$

where $\|\cdot\|$ represents the L_2 norm, and p_{i1}^{proj} denotes the orthogonal projection point of p_{i1} on s_{i2} .

Definition 4 (*Point to trajectory distance*): Point to trajectory distance (PTD) is defined as the minimum distance among the distances from point p_{i1} to all trajectory segments of T_2 .

$$d_{PTD}(p_{i1}, T_2) = \min_{s_{i2} \in T_{s2}} d_{psd}(p_{i1}, s_{i2}) \quad (2)$$

Definition 5 (*Projection point*): A projection point (say p_{i1}^2) denotes the orthogonal projection point of a point (say p_{i1}) on one segment in T_{s2} , that minimizes the distance from p_{i1} to T_2 .

3.2. Spatial proximity between trajectories

Spatial proximity is the primary consideration when measuring trajectory similarity. Real-world trajectories such as taxi trajectories and animal migration tracks typically exhibit complex geometries and varying lengths. Intrusive, noise points may exist, thereby affecting the accuracy of the trajectory similarity assessment. Therefore, we propose an improved shape-based metric to measure the spatial proximity of trajectories based on the Segment-Path Distance (Besse *et al.* 2016).

Definition 6 (*Segment-Path distance*): Segment-Path distance (SPD) is the average distance from all trajectory points in T_1 to T_2 . As shown in Figure 1, the SPD can be calculated as follows:

$$d_{SPD}(T_1, T_2) = \frac{1}{n_1} \sum_{p_{i1} \in T_1} d_{PTD}(p_{i1}, T_2) \quad (3)$$

where n_1 is the number of sampling points in T_1 and $d_{PTD}(p_{i1}, T_2)$ is the distance from p_{i1} in T_1 to T_2 .

Similarly, the SPD from T_2 to T_1 can be expressed as $d_{SPD}(T_1, T_2)$. It can be seen from Equation (3) that the SPD is a shape-based measure and could weaken the influence of noise points. However, this method fails to satisfy the symmetry requirement of a metric. Therefore, an improved shape-based similarity measure is proposed to assess the spatial similarity when T_1 and T_2 are of different lengths or T_1 is a sub-trajectory of T_2 . It is defined as the minimum distance between $d_{SPD}(T_1, T_2)$ and $d_{SPD}(T_2, T_1)$.

$$D_s(T_1, T_2) = \min(d_{SPD}(T_1, T_2), d_{SPD}(T_2, T_1)) \quad (4)$$

As shown in Equation (4), D_s is symmetric and $D_s(T_1, T_2)$ equals $D_s(T_2, T_1)$. If T_1 is a sub-trajectory of T_2 , $D_s(T_1, T_2)$ is equal to zero. Suppose T_1 is a common sub-trajectory of T_2 . In this case, the proposed distance metric can also effectively assess the similarity between T_1 and T_2 .

3.3. Movement direction consistency between trajectories

The trajectory records the movement of an object in space. Thus, to measure the similarity of the trajectories of different moving objects, the moving characteristics (e.g. moving directions) of the objects must be considered. Building on earlier definitions, this study presents a new measure that focuses on the directional similarity between trajectories. The idea is to utilize the direction consistency between the sampling points (e.g. p_i) in T_1 and their projection points (e.g. q_i) in T_2 , as shown in Figure 2.

To measure the moving direction consistency of the trajectories, vehicle heading directions are initially derived from the trajectory data. If the vehicle heading directions are not available, the moving directions at the sampling points are estimated according to the following criteria: For three adjacent sampling points, $p_i = (x_i, y_i)$, $p_{i-1} = (x_{i-1}, y_{i-1})$, and $p_{i+1} = (x_{i+1}, y_{i+1})$ in trajectory T_1 with n sampling points, the average direction of vectors from p_i to p_{i+1} and p_{i-1} to p_i is taken as the moving direction of p_i , denoted as $(\delta x_i, \delta y_i)$. Then, the moving directions of sampling points in T_1 are represented as $\{(\delta x_1, \delta y_1), \dots, (\delta x_n, \delta y_n)\}$, shown as green arrows in Figure 2.

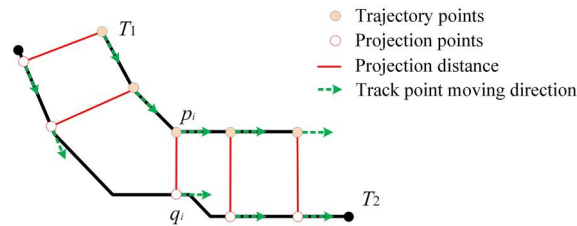


Figure 2. Illustration of the calculation of direction similarity.

Similarly, the moving directions of the corresponding projection points of T_1 on T_2 are represented by $\{(\delta x_{(1)}, \delta y_{(1)}), \dots, (\delta x_{(n)}, \delta y_{(n)})\}$. The average value of the pairwise cosine similarity between $(\delta x_i, \delta y_i)$ and $(\delta x_{(i)}, \delta y_{(i)})$ ($i = 1, 2, \dots, n$) is used to measure the moving direction difference from T_1 to T_2 , denoted as $d_{TR}(T_1, T_2)$:

$$d_{TR}(T_1, T_2) = \frac{1}{n} \sum_{i=1}^n \left(1 - \frac{\delta x_i \delta x_{(i)} + \delta y_i \delta y_{(i)}}{\sqrt{\delta x_i^2 + \delta y_i^2} \times \sqrt{\delta x_{(i)}^2 + \delta y_{(i)}^2}} \right) \quad (5)$$

Similarly, the difference in moving direction from T_2 to T_1 can be computed and denoted as $d_{TR}(T_2, T_1)$. The movement direction consistency of T_1 and T_2 is defined as the minimum value of $d_{TR}(T_1, T_2)$ and $d_{TR}(T_2, T_1)$, as shown in Equation (6). $D_r(T_1, T_2)$ is symmetric, ranging from 0 to 2. The smaller $D_r(T_1, T_2)$ is, the more similar T_1 and T_2 are to their moving directions.

$$D_r(T_1, T_2) = \min\{d_{TR}(T_1, T_2), d_{TR}(T_2, T_1)\} \quad (6)$$

3.4. Trajectory similarity—DSPD distance

After defining the spatial proximity and movement direction consistency measures of T_1 and T_2 , a comprehensive measure is proposed that integrates spatial proximity and movement direction consistency. Because spatial proximity is the dominant feature in determining the similarity of T_1 and T_2 and the dimensions of spatial distance and directional similarity are different, the movement direction consistency measure is used as a weight attached to the spatial proximity measure. The proposed directed segment path distance (DSPD) can be calculated as:

$$DSPD(T_1, T_2) = \frac{2}{2 - D_r(T_1, T_2)} \cdot D_s(T_1, T_2) \quad (7)$$

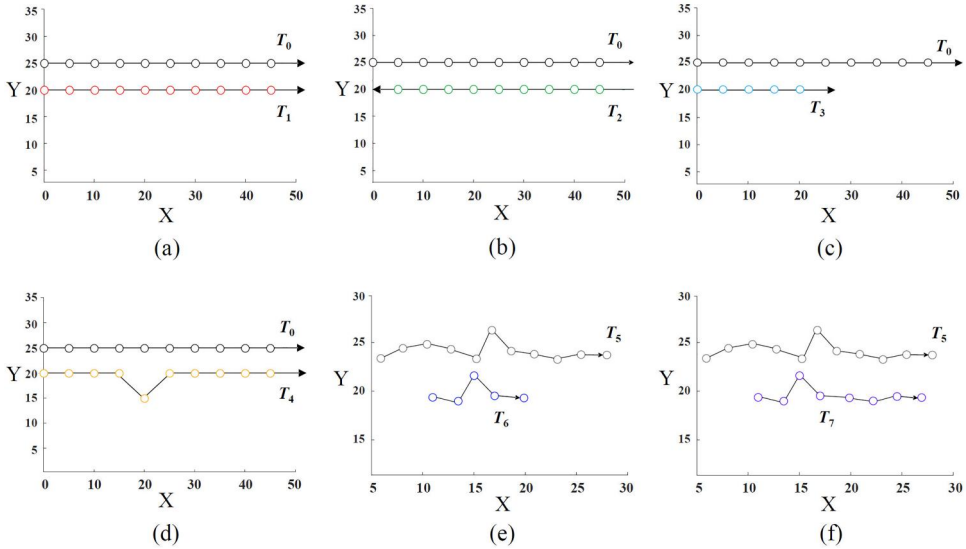


Figure 3. Six pairs of simulated trajectories. X and Y indicate the coordinates of the trajectory points.

where $2/(2-D_r(T_1, T_2))$ (say w) denotes the rectified weight of the direction consistency to spatial proximity. It decreases monotonically with the increase of $D_r(T_1, T_2)$. If the moving directions of T_1 and T_2 are identical, that is, $D_r(T_1, T_2) = 0$, then $w = 1$. This indicates that the trajectory similarity is determined only by spatial proximity. Conversely, if the moving directions of T_1 and T_2 are opposite (i.e. $D_r(T_1, T_2) = 2$), then w trends to infinity. In this case, T_1 and T_2 are completely dissimilar. The more inconsistent the trajectory directions, the greater the change in the rectified direction similarity, and the less similar the trajectories. While the average direction inconsistency is within 60° , the rectified direction similarity change is not drastic, which illustrates that slight fluctuations or noise do not substantially influence the overall similarity measurement of the trajectories. The DSPD is symmetric; the proof can be found in the [Supplementary Materials](#).

4. Experiments on simulated datasets

4.1. Simulated datasets

Six simulated trajectory datasets were used to evaluate the performance of the DSPD. The first simulated dataset contains pairs of trajectories with different lengths, directions, and noise, as shown in [Figure 3](#). T_1 has the same direction as T_0 without noise, whereas T_0 and T_2 have opposite directions. T_3 has the same direction as T_0 but is half the length. T_4 has the same direction as T_0 but contains noise. T_6 and T_7 are sub-trajectories of T_5 where T_7 extends T_6 . Compared with T_0 – T_3 , T_5 – T_6 and T_5 – T_7 also incorporate noise and directional perturbations.

The other five trajectory datasets are provided by Morris and Trivedi (2009) and are often used as benchmark datasets for evaluating the clustering of trajectories. In [Figure 4](#), these datasets cover various data scenarios, allowing the testing of the adaptability of the DSPD to trajectories of different lengths, directions, and noise.

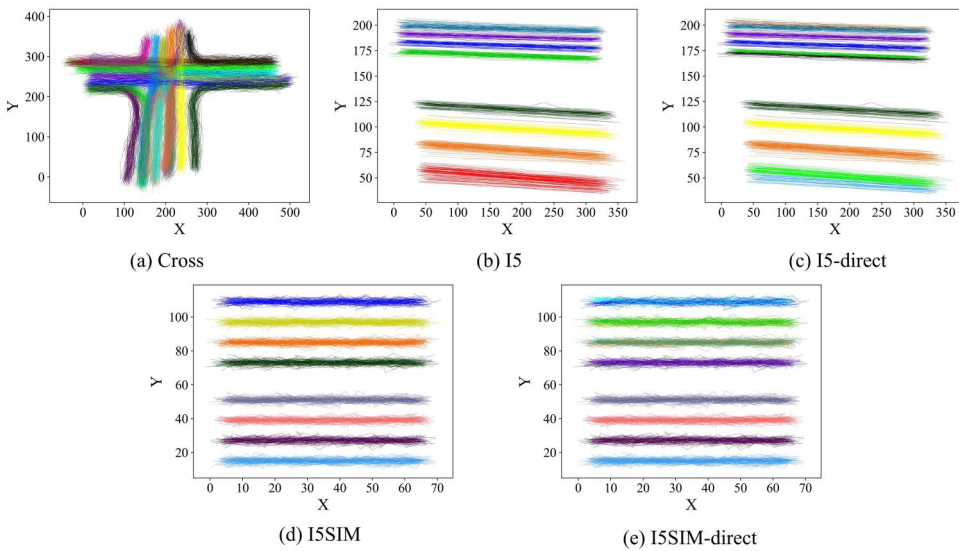


Figure 4. Simulated trajectory datasets (different colors represent different trajectory clusters).

Among them, the I5-direct and I5SIM-direct datasets are modified from the I5 and I5SIM datasets, respectively, incorporating trajectories with distinct directions.

4.2. Baselines and evaluation indicators

To evaluate the effectiveness of DSPD, we compared the results of eleven state-of-the-art trajectory similarity measures, including ED (Zhang *et al.* 2006), DTW (Berndt and Clifford 1994), HD (Chen *et al.* 2011), LCSS (Yuan *et al.* 2017), DF (Lyu *et al.* 2022), SOWD (Lin and Su 2008), SSPD (Besse *et al.* 2016), SSPD* (Zhang and Shi 2021), LCST (Deng *et al.* 2018), EDR (Chen *et al.* 2005), and VSEM (Yang *et al.* 2018). To use the ED and VSEM, we interpolated the pairs of trajectories to ensure that they had the same number of points. For the LCSS, LCST, and EDR measures, which require a distance threshold (ϵ), we tested these measures with ϵ set to 2, 5, 10, 15, 25, and 30 m, and selected the best results for comparison. The VSEM measure requires two parameters, w_1 and w_2 to weigh the spatial distance and angular similarity, respectively, and a parameter D to normalize the distances between trajectories. In the experiments, we evaluated the VSEM measure using different parameter settings. The implementation codes for these baseline measures, excluding ED, SSPD*, LCST, and VSEM, were obtained from the public Trajectory Distance Toolkit (Besse *et al.* 2016).

We employed the adjusted rand index (ARI) (Hubert and Arabie 1985) and adjusted mutual information (AMI) (Vinh *et al.* 2009) to evaluate these measures quantitatively. Both ARI and AMI generate scores ranging from -1 and 1 , where a higher value indicates a better clustering result.

Let $\mathbf{C}$ and $\mathbf{G}$ be the clustering results and ground truth, respectively. The ARI score between $\mathbf{C}$ and $\mathbf{G}$ is defined as

$$\text{ARI}(\mathbf{C}, \mathbf{G}) = \frac{2(ad - bc)}{(a + b)(b + d) + (a + c)(c + d)} \quad (8)$$

where a represents the number of trajectory pairs (e.g. T_i and T_j) that T_i and T_j belong to different clusters in $\mathbf{C}$ and $\mathbf{G}$, b represents the number of trajectory pairs that T_i and

Table 2. Results by different trajectory similarity measures on six pairwise trajectories.

Measures	T_0-T_1	T_0-T_2	T_0-T_3	T_0-T_4	T_5-T_6	T_5-T_7
ED	5.00	117.09	14.04	5.45	6.40	5.70
HD	5.00	5.00	25.50	10.00	9.43	8.94
DF	5.00	246.05	25.50	10.00	9.43	8.94
LCSS ($\epsilon = 4.9$)	0.00	0.00	0.00	0.00	0.50	0.63
LCSS ($\epsilon = 5.4$)	1.00	0.09	1.00	0.91	1.00	1.00
LCST ($\epsilon = 4.9$)	0.50	0.00	0.50	0.47	0.68	0.72
LCST ($\epsilon = 5.4$)	1.00	0.05	1.00	0.93	0.93	0.91
DTW	5.00	28.20	10.00	5.50	5.83	5.08
EDR ($\epsilon = 4.9$)	0.00	0.00	0.45	0.00	0.64	0.82
EDR ($\epsilon = 5.1$)	1.0	0.09	0.55	0.91	0.82	1.00
SSPD	6.13	15.17	9.70	6.37	4.86	4.38
SSPD*	5.00	22.25	8.59	5.31	5.24	4.62
SOWD	5.63	5.63	7.67	5.98	5.15	4.16
VSEM ($w_1=0.5, w_2=0.5, D=10$)	0.68	0.23	0.56	0.63	0.57	0.59
VSEM ($w_1=0.5, w_2=0.5, D=4$)	0.68	0.23	0.56	0.63	0.53	0.55
VSEM ($w_1=0.7, w_2=0.3, D=10$)	0.68	0.23	0.56	0.63	0.63	0.66
VSEM ($w_1=0.7, w_2=0.3, D=4$)	0.68	0.23	0.56	0.63	0.41	0.44
DSPD	5.00	$+\infty$	5.00	5.30	4.29	4.25

T_j belong to the same cluster in $\mathbf{C}$ but in different clusters in $\mathbf{G}$, c represents the number of trajectory pairs that T_i and T_j belong to the same cluster in $\mathbf{G}$ but in different clusters in $\mathbf{C}$, and d represents the number of trajectory pairs that T_i and T_j belong to the same clusters in both $\mathbf{C}$ and $\mathbf{G}$.

The AMI score between $\mathbf{C}$ and $\mathbf{G}$ is represented as:

$$AMI(C, G) = \frac{I(C, G) - E[I(C, G)]}{1/2(H(C) + H(G)) - E[I(C, G)]} \quad (9)$$

where $I(\mathbf{C}, \mathbf{G})$ denotes the mutual information between $\mathbf{C}$ and $\mathbf{G}$, $E(MI)$ represents the expected value of I under random clustering, $H(\mathbf{C})$ is the entropy of $\mathbf{C}$, and $H(\mathbf{G})$ is the entropy of $\mathbf{G}$.

4.3. Comparison of different measures on six trajectory pairs

The results of the DSPD and baseline measures for the six simulated trajectory pairs are listed in Table 2. This shows that DSPD is effective in distinguishing trajectories in distinct directions (e.g. T_0-T_2), mitigating the influence of noise (e.g. T_0-T_4), and handling trajectories with unequal lengths, such as sub-trajectory scenarios (i.e. T_0-T_3 , T_5-T_6 , and T_5-T_7). The LCSS, LCST, and EDR are similar to the DSPD in these aspects but are affected by the distance threshold, whereas the DSPD is parameter-free. Choosing an appropriate threshold can sometimes be challenging and a global threshold may not satisfy the requirements. Hausdorff and Euclidean distances make it difficult to distinguish trajectories in different directions (e.g. T_0-T_2). They are sensitive to noise (e.g. T_0-T_4) and exhibit bias for misaligned trajectories (e.g. T_0-T_4 , T_5-T_6 , and T_5-T_7). The DF and DTW measures are also sensitive to noise. For trajectory pairs with slight changes in direction (e.g. T_0-T_1 , T_0-T_2 , T_0-T_3 , and T_0-T_4), VSEM is stable and insensitive to the parameters; otherwise, it may differ with changes in parameters w_1 , w_2 , and D , e.g. T_5-T_6 and T_5-T_7 . Overall, the results obtained by DSPD are more in line with human expectations, and it performs better on the simulated data than the other methods.

4.4. Comparison of different measures for trajectory clustering on simulated datasets

We evaluated DSPD and 11 baseline measures by applying them to two typical clustering algorithms: i.e. hierarchical agglomerative clustering (Vijaya *et al.* 2019), and the density-based spatial clustering of applications with noise (HDBSCAN) (Wang *et al.* 2021). The number of clusters in the hierarchical clustering was set to the ground truth, and the minimum size of the clusters in the HDBSCAN algorithm was determined experimentally. The effectiveness of the trajectory similarity measures was reflected in the quality of the clustering results.

Table 3 presents the accuracy of the clustering results obtained using the hierarchical agglomerative clustering algorithm with different trajectory similarity measures. The best results are highlighted in bold and underlined. When the optimal parameters of the LCST and LCSS were selected, the hierarchical clustering algorithm using the LCST and LCSS yielded better clustering results than the other measures on the

Table 3. ARI and AMI scores of the clustering results obtained by the hierarchical agglomerative clustering algorithm with different trajectory similarity measures.

Evaluation indicator	Measure	Cross	I5	I5SIM	I5-direct	I5SIM-direct	Average score
ARI	ED	0.9868	0.6536	1.0000	0.7366	1.0000	0.8754
	HD	0.9869	0.6000	1.0000	0.5608	0.7813	0.7858
	DF	0.9869	0.3738	1.0000	0.4791	1.0000	0.7680
	LCSS	0.9806	0.9846	1.0000	0.9471	1.0000	0.9825
		(eps = 25)	(eps = 10)	(eps = 20)	(eps = 5)	(eps = 20)	
	LCSS	0.8850	0.9344	1.0000	0.5511	1.0000	0.8741
		(eps = 5)	(eps = 5)	(eps = 25)	(eps = 2)	(eps25)	
	LCST	0.9816	0.9950	1.0000	0.9883	1.0000	0.9930
		(eps = 25)	(eps = 25)	(eps = 20)	(eps = 10)	(eps = 25)	
	LCST	0.8723	0.8301	0.7335	0.5869	0.8289	0.7703
		(eps = 5)	(eps = 5)	(eps = 30)	(eps = 2)	(eps = 30)	
	DTW	0.7894	0.6973	1.0000	0.6116	1.0000	0.8197
	EDR	0.4842	0.6748	1.0000	0.5680	1.0000	0.7454
	SSPD	0.9813	0.9641	1.0000	0.8514	0.7678	0.9129
	SSPD*	0.8438	0.6295	1.0000	0.6676	1.0000	0.8282
	SOWD	0.9553	0.9562	1.0000	0.8667	0.7593	0.9075
	VSEM	0.9856	0.7354	1.0000	0.7580	1.0000	0.8958
	DSPD	0.9641	0.9898	1.0000	0.9884	1.0000	0.9885
AMI	ED	0.9897	0.7652	1.0000	0.8381	1.0000	0.9186
	HD	0.9902	0.7357	1.0000	0.7238	0.8714	0.8642
	DF	0.9905	0.5594	1.0000	0.7082	1.0000	0.8516
	LCSS	0.9855	0.9846	1.0000	0.9575	1.0000	0.9855
		(eps = 25)	(eps = 10)	(eps = 20)	(eps = 5)	(eps = 20)	
	LCSS	0.9256	0.9483	1.0000	0.7750	1.0000	0.9298
		(eps = 5)	(eps = 5)	(eps = 25)	(eps = 2)	(eps = 25)	
	LCST	0.9867	0.9933	1.0000	0.9888	1.0000	0.9938
		(eps = 25)	(eps = 25)	(eps = 20)	(eps = 10)	(eps = 25)	
	LCST	0.9497	0.9006	0.8512	0.8124	0.9272	0.8882
		(eps = 5)	(eps = 5)	(eps = 30)	(eps = 2)	(eps = 30)	
	DTW	0.9111	0.7990	1.0000	0.7838	1.0000	0.8988
	EDR	0.7470	0.7756	1.0000	0.7281	1.0000	0.8501
	SSPD	0.9853	0.9616	1.0000	0.8985	0.8675	0.9426
	SSPD*	0.9273	0.7199	1.0000	0.7826	1.0000	0.8860
	SOWD	0.9758	0.9531	1.0000	0.9053	0.8651	0.9399
	VSEM	0.9882	0.8172	1.0000	0.8337	1.0000	0.9278
	DSPD	0.9820	0.9874	1.0000	0.9889	1.0000	0.9917

simulated datasets. However, the accuracies of the LCSS and LCST decreased significantly with different parameter values. The best results obtained by the LCSS and LCST on different datasets used different distance thresholds. This may limit their practicability on real-world datasets, particularly for trajectory data with heterogeneous densities and without prior knowledge. The proposed DSPD achieved results comparable to those of the LCSS and LCST and did not require additional parameters. Most trajectory similarity measures performed well on the I5SIM and I5SIM-directed datasets. This may be owing to the simplicity of the I5SIM dataset, where distinct clusters make it suitable for almost all trajectory similarity measures. The HD, SSPD, and SOWD on the I5SIM-direct and I5-direct datasets performed poorly because they could not measure the moving direction difference of the trajectories.

Table 4 presents the clustering results obtained using the HDBSCAN with different trajectory similarity measures. The results indicate that the DSPD is the most effective measure overall, followed by the LCSS, in terms of the average accuracy on all datasets. However, the HD, DF, DTW, EDR, and SOWD measures were less effective,

Table 4. ARI and AMI scores of the clustering results obtained by the HDBSCAN with different trajectory similarity measures.

Evaluation indicator	Measure	Cross	I5	I5SIM	I5-direct	I5SIM-direct	Average score
ARI	ED	0.7414	0.4750	1.0000	0.5519	1.0000	0.7537
	HD	0.6267	0.4522	1.0000	0.3752	0.8764	0.6661
	DF	0.7414	0.2482	1.0000	0.3028	1.0000	0.6585
	LCSS	0.7414	0.9664	1.0000	0.9029	1.0000	0.9221
	LCST	0.6026	0.9672	1.0000	0.9529	1.0000	0.9045
	DTW	0.7414	0.3749	1.0000	<u>0.4986</u>	1.0000	0.7230
	EDR	0.4819	0.2489	1.0000	0.3327	1.0000	0.6127
	SSPD	0.6262	0.8612	1.0000	0.7506	0.8764	0.8229
	SSPD*	0.7414	0.8378	1.0000	0.8626	1.0000	0.8884
	SOWD	0.5801	0.4729	1.0000	0.3787	0.8764	0.6616
	VSEM	0.6267	0.4686	1.0000	0.5398	1.0000	0.7270
	DSPD	0.7753	0.9867	1.0000	0.9235	1.0000	0.9371
AMI	ED	0.9179	0.6815	1.0000	0.7623	1.0000	0.8723
	HD	0.8877	0.6593	1.0000	0.6208	0.9334	0.8202
	DF	0.9179	0.4639	1.0000	0.5933	1.0000	0.7950
	LCSS	0.9179	0.9569	1.0000	0.9374	1.0000	0.9624
	LCST	0.8745	0.9650	1.0000	0.9633	1.0000	0.9606
	DTW	0.9179	0.6704	1.0000	<u>0.7933</u>	1.0000	0.8763
	EDR	0.7897	0.4592	1.0000	0.5692	1.0000	0.7636
	SSPD	0.8866	0.8822	1.0000	0.8242	0.9334	0.9053
	SSPD*	0.9179	0.8634	1.0000	0.8995	1.0000	0.9362
	SOWD	0.866	0.7646	1.0000	0.7094	0.9334	0.8547
	VSEM	0.8877	0.6956	1.0000	0.8233	1.0000	0.8813
	DSPD	0.9318	0.9824	1.0000	0.9459	1.0000	0.9720

particularly on the I5 and I5-direct datasets. The main limitation of HD, SSPD, and SOWD is the inaccurate measurement of trajectories with distinct directions. The LCSS, LCST, and EDR are limited by the determination of optimal parameter thresholds. A different parameter setting may lead to a significant decrease in the clustering quality.

Overall, the above experiments demonstrated the superiority of the DSPD over other measures for finding similar trajectories and that it can be used in different clustering algorithms for detecting diverse clustering patterns, especially for density-based methods (e.g. HDBSCAN). When the ground truth of the cluster numbers is known, the hierarchical clustering algorithm can obtain better clustering results than HDBSCAN on these simulated datasets. The main reason is that HDBSCAN still has difficulty in separating weakly connected clusters and may wrongly classify some trajectories at cluster boundaries as noise.

4.5. Complexity analysis

Given one trajectory T_m with m trajectory points and another trajectory T_n with n trajectory points, the time complexity of the proposed DSPD is $O(nm)$. The time complexity aligns with most baselines except for the Euclidean distance ($O(n+m)$) and Fréchet distance ($O(nm\log(nm))$). Although DSPD has the same theoretical time complexity as many baselines, the actual running times vary owing to implementation differences. For example, calculating the Hausdorff distance involves determining the maximum of the minimum distances from all points in T_m to T_n , which is less computational than SSPD which requires finding projection points for points in T_m on T_n . The DSPD needs

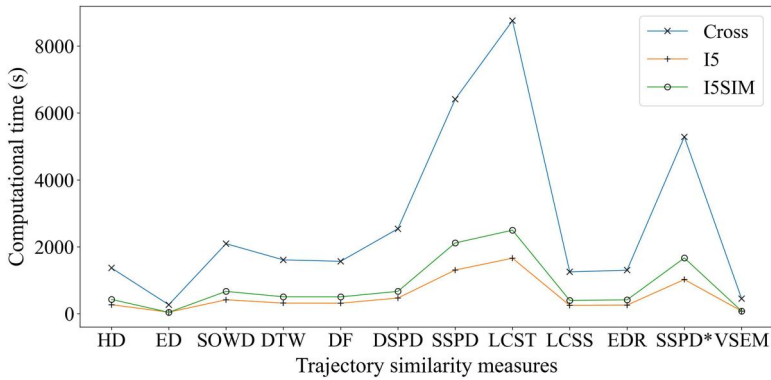


Figure 5. Running times of different trajectory similarity measures on the Cross dataset (with 1900 trajectories), the I5 dataset (with 806 trajectories), and the I5SIM dataset (with 800 trajectories).

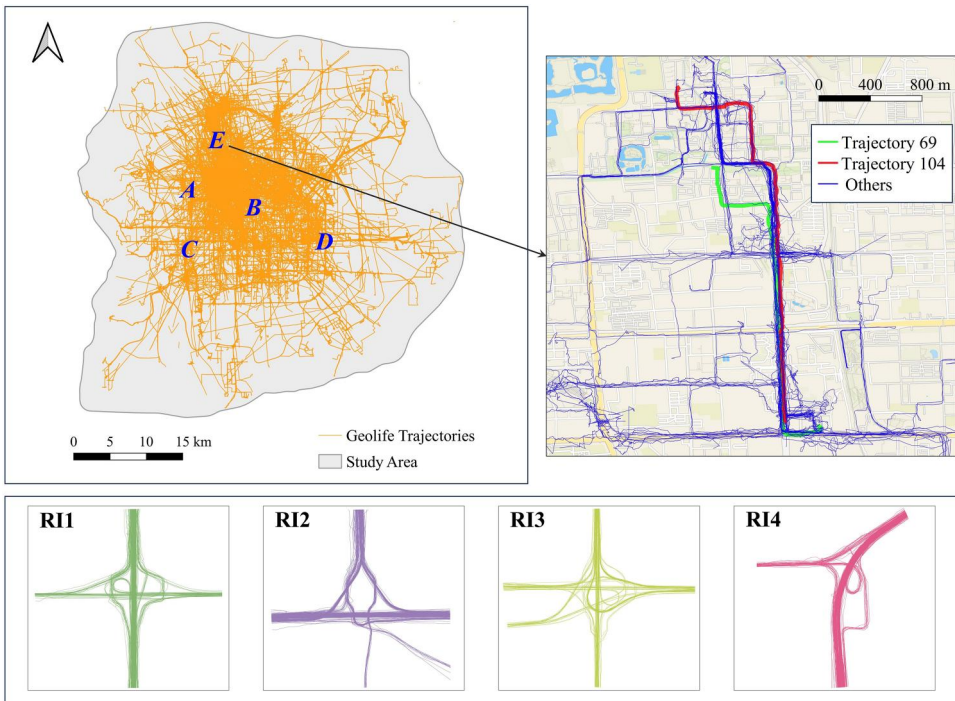


Figure 6. Study area and Geolife dataset (only trajectories within Sixth Ring Road are shown). Trajectories in regions A, B, C, and D are extracted to generate RI1, RI2, RI3, and RI4 datasets; Trajectories in region E are extracted and used in the trajectory retrieval experiment.

to calculate the direction consistency, and therefore theoretically takes more time than the SSPD. To enhance the efficiency of DSPD, our implementation optimizes the calculation of the projection distances by identifying the point in the other trajectory nearest to the given point. This limits the projection distance calculations to trajectory segments surrounding the nearest point and thus significantly reduces the running time of the DSPD.

We also counted the running time of the proposed DSPD and baseline measures on the Cross, I5, and I5SIM datasets to further compare the computational efficiency, as shown in Figure 5. In the experiments, the DSPD and baseline measures were implemented in Python 3.8 and tested on a desktop computer with AMD R7 5800X CPUs at 3.80 GHz and 64 GB of RAM.

5. Applications of DSPD

To evaluate the practicality of the DSPD, the Geolife trajectory dataset obtained from Microsoft Research Asia (Zheng *et al.* 2010) was used for two case studies. This dataset consists of trajectories from 182 users engaged in various transportation modes and daily activities, such as bus commuting, walking, shopping, and biking. The data were collected using multiple GPS devices containing trajectories of different lengths. The trajectories used in the case studies are shown in Figure 6.

5.1. Clustering of similar trajectories for road network generation

5.1.1. Experimental design and data description

In the first case study, trajectories in four intersection regions **A**, **B**, **C**, and **D** were selected to group trajectories on different roads, and roads were generated based on the clustering results. The trajectory data in four intersection regions are denoted as RI1, RI2, RI3, and RI4. As shown in Figure 6, they contain diverse traffic modes, which allow us to verify the effectiveness of DSPD by grouping the trajectories in different roads into different clusters. We manually labeled the trajectories by overlapping the trajectory data on the online road network map to establish the ground truth of the trajectory clusters.

5.1.2. Trajectory clustering results and road generation

We used HDBSCAN as the clustering algorithm because the ground truth of the cluster numbers is usually unknown in advance, and compared with other clustering

Table 5. Clustering results of four datasets using the DSPD, Hausdorff distance, and LCST.

Data	Parameter	Similarity measure	ARI	AMI	SIC	Detected cluster number	Ground truth
RI1	minimum cluster size = 3	DSPD	0.9985	0.9957	0.8367	12	12
		HD	0.5719	0.8403	0.8259	11	
		LCST (<i>eps</i> = 20)	0.9901	0.9650	0.5025	12	
		LCST (<i>eps</i> = 25)	0.9930	0.9734	0.5535	11	
RI2	minimum cluster size = 3	DSPD	0.9973	0.9964	0.9057	10	10
		HD	0.7335	0.9036	0.8940	9	
		LCST (<i>eps</i> = 10)	0.9725	0.9584	0.3866	7	
		LCST (<i>eps</i> = 15)	0.9842	0.9596	0.4741	8	
RI3	minimum cluster size = 2	DSPD	1.0000	1.0000	0.9123	18	18
		HD	0.8181	0.9292	0.9029	15	
		LCST (<i>eps</i> = 10)	0.9404	0.9220	0.3104	14	
		LCST (<i>eps</i> = 15)	0.9763	0.9729	0.4019	16	
RI4	minimum cluster size = 3	DSPD	0.9960	0.9941	0.7915	9	9
		HD	0.4581	0.7561	0.8918	8	
		LCST (<i>eps</i> = 10)	0.9811	0.9429	0.3132	8	
		LCST (<i>eps</i> = 15)	0.9941	0.9730	0.4216	9	

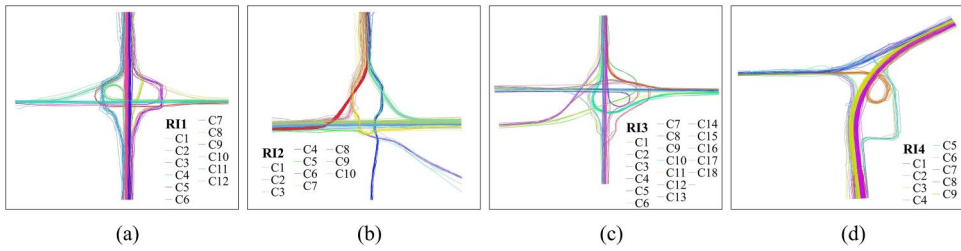


Figure 7. Clustering results of (a) RI1, (b) RI2, (c) RI3, and (d) RI4 data with the DSPD. Different colors in the subfigures represent different clusters.

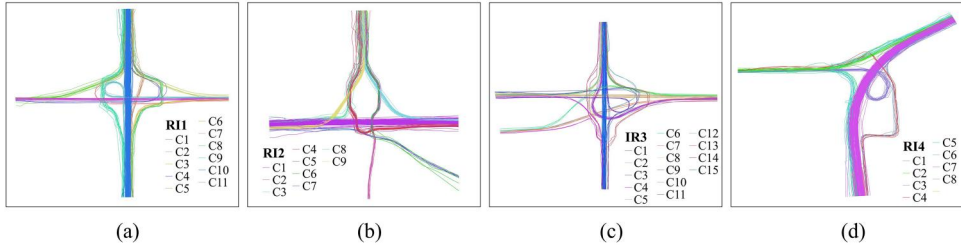
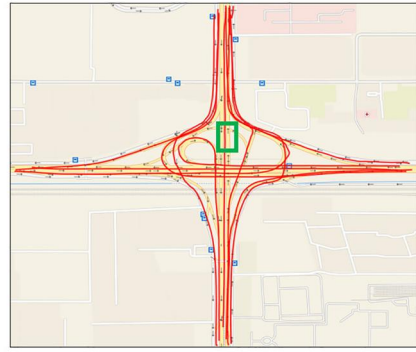
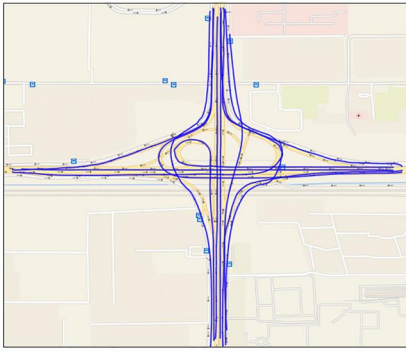


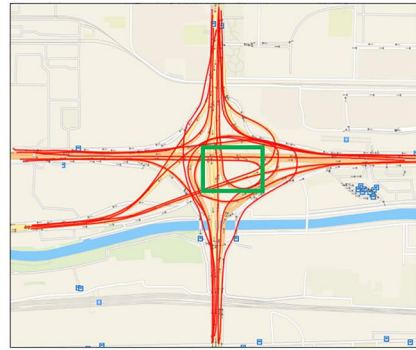
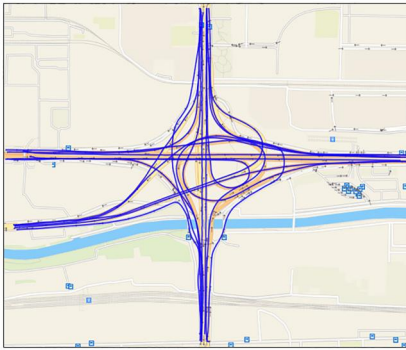
Figure 8. Clustering results of (a) RI1, (b) RI2, (c) RI3, and (d) RI4 data with the Hausdorff distance. Different colors in the subfigures represent different clusters.

algorithms, HDBSCAN requires few parameters. Using the same clustering parameter settings, the clustering results of HDBSCAN with DSPD were compared with those based on Hausdorff distance and LCST. The ARI, AMI, and Silhouette Coefficient (SIC) (Rousseeuw 1987) indicators were used for performance evaluation. The results in Table 5 demonstrate that the DSPD outperformed the Hausdorff distance and LCST in clustering real-world trajectories. Applying the DSPD enabled the HDBSCAN to group trajectories into coherent clusters more accurately. The number of clusters obtained using the DSPD was closer to the ground truth values. Although clustering using the Hausdorff distance had a higher SIC score for the RI4 dataset, this did not indicate an overall better performance owing to the limitation of SIC relating to evaluating non-spherical clusters. As shown in Figures 7 and 8, clustering using the Hausdorff distance could not distinguish trajectories with different directions (i.e. trajectories on lanes with different directions), whereas clustering using the DSPD identified clusters with distinct directions. As shown in Table 5, the results of clustering using the LCST differ with different distance thresholds (ϵ s). For the same trajectory data, the optimal results using the LCST for different road intersections were obtained under different ϵ s values. The difficulty in setting the parameters limits the power of the LCST in practical applications.

Based on the clustering results using the DSPD and the Hausdorff distance, we generated the centerlines of the road lanes using the soft k-segments algorithm (Verbeek *et al.* 2001). The extracted road centerlines illustrate that clustering using the DSPD can more accurately divide trajectories in road intersection regions into different clusters corresponding to the underlying road lanes. In contrast, clustering using the Hausdorff distance cannot distinguish the trajectories that have opposing directions on road lanes with close spatial proximity well, as shown in the green boxes in



(a) Generated road travel paths in RI1 (left: DSPD, right: Hausdorff)



(b) Generated road travel paths in RI3 (left: DSPD, right: Hausdorff)

Figure 9. The extracted road travel paths using the proposed DSPD and Hausdorff distance.

Figure 9. The results show that the DSPD has great potential for generating lane-level road networks from high-precision trajectory data.

5.2. Common trajectory retrieval for ride-sharing path planning

5.2.1. Experimental design and data description

Recognizing common travel paths of crowds is of great value for human mobility pattern detection and promotes the efficiency of ride-sharing path planning in intelligent transportation services. In this case study, given a user's trajectory, we aim to find the top k most similar trajectories from the trajectories of other users. We selected 126 trajectories from the Geolife dataset, as shown in Figure 6. Each trajectory was assigned a unique ID (from 1 to 126), and the raw trajectories were smoothed using moving averages (the window size was set to 30 m) to reduce the noise points caused by GPS positioning errors.

5.2.2. Crowd common travel path recognition

We applied the DSPD, LCSS, LCST, and the Hausdorff distance to search for the most similar trajectories (i.e. common travel paths) to the given trajectory. Two distance thresholds were used for LCSS and LCST (i.e. 50 and 200 m). The top ten most similar

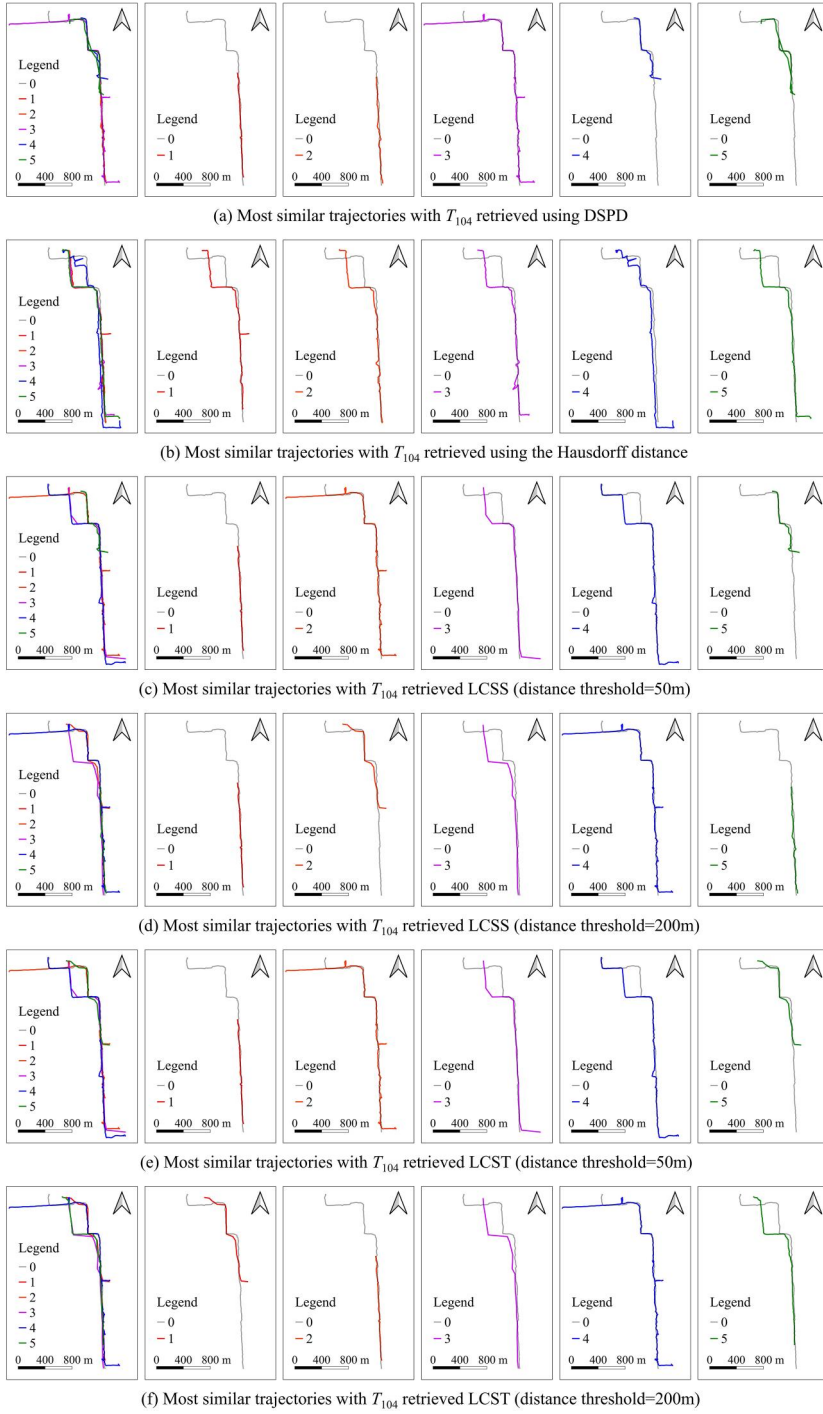


Figure 10. The five retrieved trajectories most similar to T_{104} (labeled as 0) using different similarity measures. The numbers 1–5 in the legend represent trajectories ranked in decreasing order of similarity to T_{104} .

trajectories retrieved using different trajectory similarity measures were different. The results obtained using the LCSS and LCST also differed under different *eps* values.

Taking the source trajectory with ID 104 (hereafter referred to as T_{104}) as an example, we visualized T_{104} and retrieved similar trajectories. The retrieved trajectories are sorted in descending order according to their similarity. As shown in Figure 10a, the top five trajectories similar to T_{104} retrieved using the DSPD were relatively consistent with the manual observation results. The DSPD discovered the sub-trajectories of T_{104} . The start and end points of similar trajectories retrieved using the Hausdorff distance were close to those of T_{104} ; however, there were some local trajectory segments with large separations, as illustrated in Figure 10b. The LCSS and LCST can also identify common sub-trajectories. However, they tended to find a longer trajectory that was similar to T_{104} , regardless of the presence of large gaps between the local segments, as shown in Figure 10c–f. When the distance threshold is relatively large, the similarity of the trajectory that has segments significantly deviating from T_{104} may be overestimated; for example, the similarity of the third trajectory in Figure 10d–f. The fourth trajectory in Figure 10d–f is more similar to T_{104} although the calculated results are not consistent with our cognition. From the results in Figure 10, the DSPD generally performs better in finding similar trajectories or common movement patterns compared with the baseline measures (i.e. Hausdorff distance, LCSS, and LCST). Similar trajectories found by the proposed DSPD are more consistent with our cognition. The results demonstrate the great potential of the DSPD for common travel path recognition. It can be used in sociology, ecology, intelligent transportation systems for human mobility, species migration analysis, ride-sharing path planning, etc.

6. Sensitivity analysis

To examine the capability of DSPD in handling trajectories with different sampling frequencies and noise levels, we conducted two sensitivity analyses using a pair of similar trajectories (T_1 and T_2) from the RI2 dataset. T_1 contains 26 sampling points with an average interval distance of 47.9 m. T_2 has 94 sampling points with an average interval distance of 13.3 m. The mean distance from each point in T_1 to its nearest point in T_2 is approximately 7 m. Conversely, it is 15.2 m from T_2 to T_1 .

6.1. Sensitivity tests to different sampling frequencies

We manually removed a certain percentage of sampling points in T_1 while maintaining little change in the overall shape of the trajectory. The deletion rates were 10%, 30%, 50%, and 70%, respectively. We then calculated the trajectory distances between T_2 and reconstructed T_1 using the DSPD and baseline measures and computed the distance change rates before and after deletion. The results, as shown in Figure 11, demonstrate that the DSPD, SSPD, and SOWD measures are robust to changes in sampling rates. Conversely, the Euclidean distance, discrete Fréchet, and DTW are more sensitive to changes in sampling rates among the 11 baseline measures.

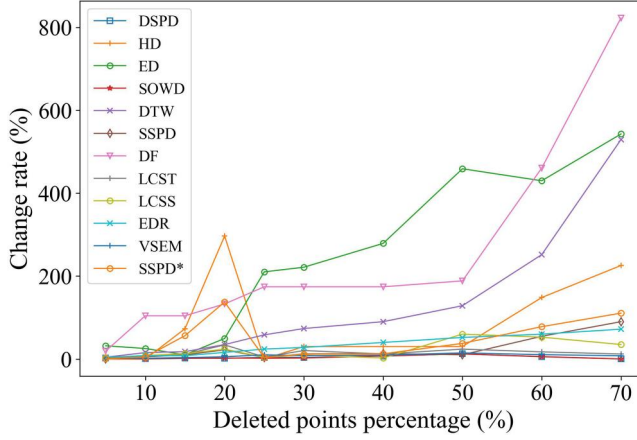


Figure 11. Distance change rates of the DSPD and baseline measures under different sampling frequencies.

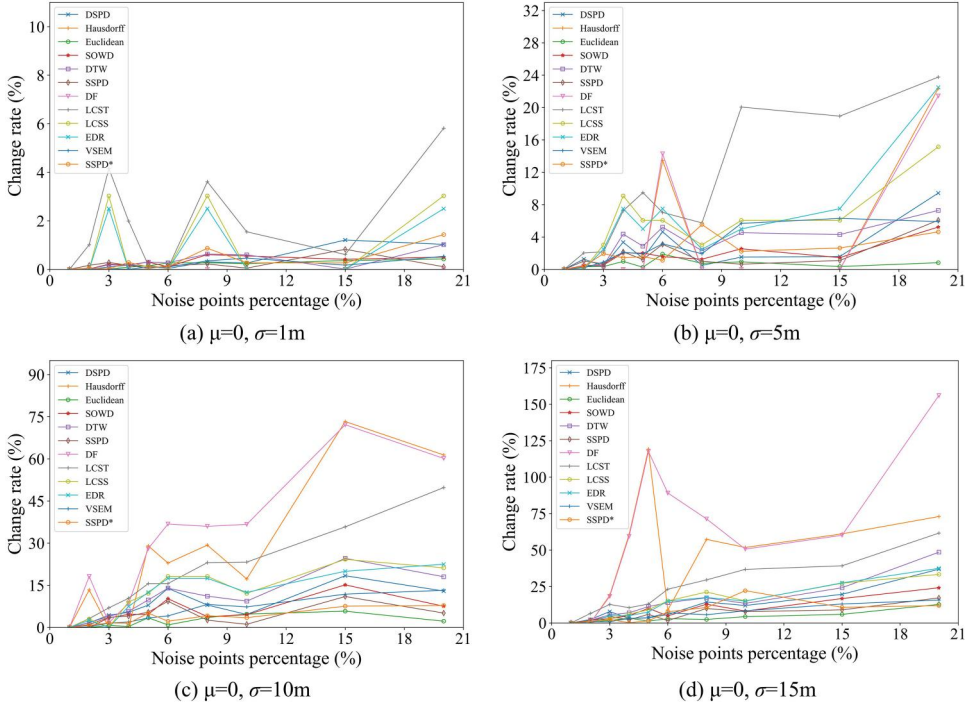


Figure 12. Distance change rates of the DSPD and baseline measures under different noise levels.

6.2. Sensitivity test to noise

We added random Gaussian noise with a mean value (μ) of 0 and standard deviation (σ) of 1, 5, 10, and 15 m, respectively to the trajectories T_1 and T_2 at different noise ratios from 1–20%. The distance thresholds for the LCSS, LCST, and EDR were set to 10 m because the shortest distances from most sampling points in T_1 to T_2 were less than 10 m. We calculated the distances of the reconstructed trajectories T_1 and T_2

under different noise ratios and measured the distance change rates before and after adding noise. As shown in [Figure 12](#), the DSPD, Euclidean, VSEM, SOWD, SSPD, and SSPD* measures are more robust to noise. In contrast, the Hausdorff and Fréchet distances change dramatically when noise is added, particularly with σ of 10 and 15 m. The LCSS, LCST, and EDR also exhibit significant changes because the noise points caused fewer matched point pairs. Besides, the addition of noise perturbs the moving directions of the sampling points in the trajectory, causing the LCST to change more than the LCSS and EDR.

The sensitivity analyses above show that shape-based measures (e.g. DSPD, SSPD, VSEM, and SOWD) are less sensitive to noise. These measures, along with the LCSS, LCST, and EDR, show greater resilience in handling trajectories of different lengths. However, the Fréchet distance, Hausdorff distance, Euclidean distance, and DTW are more susceptible when measuring the similarity of trajectories with varying lengths. And the Hausdorff distance and Fréchet distance also show high sensitivity to minor perturbances. When the deviation induced by noise surpasses the distance threshold, similarity calculations using the LCSS, LCST, and EDR measures become more skewed.

7. Conclusion and future work

This paper proposes an improved shape-based trajectory similarity measure that considers spatial proximity and direction consistency. The proposed DSPD can effectively differentiate trajectories with variations in the spatial distance and moving direction. Compared to most existing similarity measures, the DSPD is robust in measuring the similarity of crowdsourced trajectories with different lengths, noise, or small perturbances. We conducted experiments on the simulated trajectory datasets and Geolife trajectory dataset and compared the DSPD with baseline measures. The simulated experimental results show that the DSPD is more effective in measuring trajectory similarity than alternative representative measures under different scenarios. Two case studies of road path extraction and similar crowd movement detection and sensitivity analyses also indicated the effectiveness, robustness, and applicability of the DSPD. The proposed DSPD can not only find similar travel paths sharing similar traffic behaviors, but can also discover similar routes (or movement patterns) that share the same, similar, and different start or end positions from noisy trajectories. This provides a more accurate and robust trajectory similarity measurement that can facilitate movement pattern analysis and many other geographical services based on trajectories.

Although the DSPD has shown promising results, there are still two limitations to address. First, the time complexity of the DSPD may still lead to slow computational efficiency when dealing with large-scale trajectory datasets. Second, this study did not consider the time similarity of trajectories, which is crucial for some applications, such as moving cluster detection and following pattern discovery. In the future, we will continue to integrate the time similarity of trajectories to improve the DSPD. Furthermore, parallel computing strategies will be applied to increase the efficiency of the algorithm.

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Disclosure statement

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Data and codes availability statement

The source data and codes for this study are available on 'figshare.com' with the identifier at the public link: <https://doi.org/10.6084/m9.figshare.25712415>. The research applied codes for several baselines from third parties (<https://github.com/bguillouet/traj-dist>), who are not affiliated with the research.

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